

Online Resource 1

OPENER: Open Partitioning Embedding for Named Entity Recognition

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This Online Resource gathers supplementary tables and figures cited in the main article: the test-set bootstrap of the released model (Table S1), the four-axis means of every system (Fig. S1), and the zero-shot embedder comparison with Pile-NER (Table S2).

Table S1 Test-set bootstrap robustness of the released (seed-42) model on three representative datasets

Model	CoNLL-2003		CrossNER-Music		FabNER	
OPENER-ZS _{ind}	66.1 ± 1.4	[63.5, 69.0]	55.5 ± 0.9	[53.8, 57.3]	36.2 ± 0.7	[34.7, 37.5]
OPENER-Sup	71.7 ± 1.5	[68.9, 74.6]	72.4 ± 0.9	[70.6, 74.2]	59.8 ± 0.9	[57.9, 61.6]

Typing-on-gold AMI (↑) as bootstrap mean ± std [95% percentile CI], over $K=1000$ resamples of the gold mentions. The transductive refinement is excluded, as it adapts to the resampled test distribution. The bootstrap means match the full-test-set scores, and the stds stay at or below 1.5 AMI, indicating stable scores.

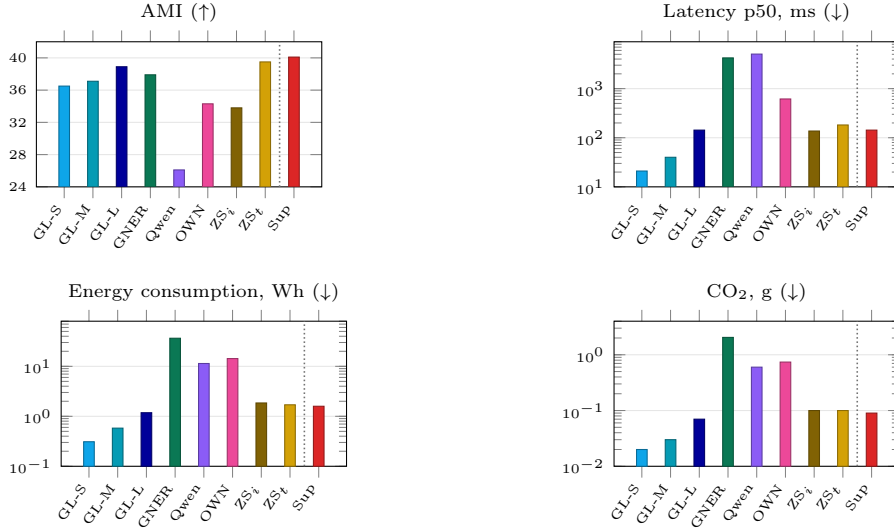


Fig. S1 Per-model means on the four axes (13-set averages, Table 7 of the main article). The light dotted line separates the *zero-shot* systems (left) from the *supervised* OPENER-Sup (right). Latency, energy and CO₂ use a log scale. Labels: GL-S/M/L = GLiNER sizes, OWN = OWNER, ZS_i = OPENER-ZS_{ind}, ZS_t = OPENER-ZS_{trans}, Sup = OPENER-Sup

Table S2 Zero-shot typing: effect of the contrastive embedder on OPENER-ZS_{ind}(typing-on-gold, 13-set means at the released seed (42))

Embedder	Training data	AMI	F ₁	Acc
hard-neg. mining (selected)	CoNLL + targets' train	44.4	46.8	53.6
Pile-NER (entity $\leftrightarrow$ label)	Pile-NER only	42.2	50.6	53.9
+ label-name hard mining	Pile-NER only	39.7	41.4	44.7

The nearest label-name centroid head is fixed, only the embedder changes. AMI, macro- F_1 , accuracy ($\times 100$, $\uparrow$). Best per column in bold. The selected hard-negative-mining embedder (target-tuned) is compared with Pile-NER variants (Pile-NER is the training corpus released with UniversalNER, cited in the main article) that are strictly domain-agnostic (never see target data). We report Pile-NER as a negative ablation (Section 4.7 of the main article).